

Lecture 19-24

Q#1: Construct backward difference table from the following table

x	1	3	5
$f(x)$	2	4	7

Q#2: Construct backward difference table from the following table

x	1	5	10
$f(x)$	3	7	14

Q#3: Construct a backward difference table from the following values of x and y .

x	1	2	3	4	5
$y=f(x)$	2	5	10	22	39

Q#4: Prove that $E^{-\frac{1}{2}} = \mu - \frac{\delta}{2}$

Q#5: Prove that $\mu\delta = \frac{\Delta + \nabla}{2}$

Q#6: Define shift operator, Average operator, Differential operator.

Q#7: Find the value of p from the following table of values if $f(18)$ is to be calculated

x	1	2	3	4	5
$f(x)$	35	39	41	53	65

Q#8:

Find the interpolating polynomial for the following data by Newton's Forward Difference Formula

x	y	Δy	$\Delta^2 y$	$\Delta^3 y$
0	1			
		2		
1	3		3	
		5		0
2	8		3	
		8		
3	16			

Q#9:

Find the value of p from the following table of values if $f(.39)$ is to be calculated

x	.4	.5	.6	.7	.8
$f(x)$	.29	.35	.37	.42	.47

Q#10:

To estimate $f(3.8)$ from the following table of values what should the value of p be?

x	1	2	3	4	5
$f(x)$	1	4	8	9	12

Q#11:

Consider the following table of values

x	2	4	6	8	10	12	14
$F(x)$	23	93	259	559	1031	1713	2643

Use Newton's backwards difference formula to compute $f(11)$ with $x_n = 10$ and $p = 0.5$

Q#12:

Evaluate $f(2.5)$ from the following data by Lagrange's interpolation formula

x	1	3	7
$f(x)$	4	18	21

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Question: Find Newton's Divided Difference table for the following data.

x	0	1	2	4	5
$f(x)$	1	9	12	5	6

Question: Find Newton's Divided Difference table for the following data.

x	0	1	3
$f(x)$	2	4	20

Question: Compute $f(4)$ for the data

x	0	2	3	6
$f(x)$	7	6	9	21

using the Lagrange's Interpolation formula.

Answer: $f(4) = \frac{118}{9}$

Question: Using the two dimensional interpolation tabulate the values of the function $f(x, y) = x^2 + y + 3y^2$ for $x=0,1,2$, and $y=0,1,2$.

Question: Find Newton's forward difference table from the following data.

x	0.0	0.1	0.2	0.3	0.4
$f(x)$	1	0.9048	0.8187	0.6408	0.4703

Question: Find $y'(1)$ from the following forward difference table.

x	y	Δy	$\Delta^2 y$	$\Delta^3 y$	$\Delta^4 y$
1.0	2				
		1.5			
1.2	3.5		-6.7		
		-5.2		15.9	
1.4	-1.7		9.2		-28.2
		4		-12.3	
1.6	2.3		-3.1		
		0.9			
1.8	3.2				

Answer: $y'(1) = 86$

Question: Find $y''(8)$ from the following backward difference table.

x	y	∇y	$\nabla^2 y$	$\nabla^3 y$	$\nabla^4 y$	$\nabla^5 y$	$\nabla^6 y$
1	2						
		7					
2	9		12				
		19		6			

3	28		18		0		
		37		6		0	
4	65		24		0		0
		61		6		0	
5	126		30		0		0
		91		6		0	
6	217		36		0		
		127		6			
7	344		42				
		169					
8	513						

Solution: $y''(8) = 48$

Question: Construct forward difference table from the following table of values.

x	1.00	1.05	1.10	1.15	1.20
y	1.0000	1.0247	1.0488	1.0724	1.0454

Question: Find first derivative of $f(x)$ at $x=0.1$ using two point equation

Where $f(x)$ is given by

x	0.1	0.2	0.3	0.4	0.5	0.6
$f(x)$	0.105	0.342	0.652	0.756	0.812	0.924

Solution: $f'(0.1) = 2.37$

Question: Find the second derivative of $f(x)$ at $x = 0.3$ using three-point equation.

Where $f(x)$ is given by

x	0.1	0.2	0.3	0.4	0.5	0.6
$f(x)$	0.125	0.302	0.602	0.706	0.812	0.924

Solution: $f''(0.3) = -19.6$

Question: If $F(h) = 244.2354$ and $F(\frac{h}{2}) = 247.1379$, then find $F_1(\frac{h}{2})$ using Richardson's extrapolation limit.

Solution: $F_1(\frac{h}{2}) = 248.1054$

Question: Find $F(h)$, using Richardson's extrapolation limit, while finding $y'(0.01)$ to the function $y = \frac{1}{x}$ with $h = 0.007$.

Solution: $F(h) = -19607.8431$